

Heritable variation drives rapid evolution of thermal performance curves in the protist *Tetrahymena thermophila*

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TITLE

Heritable variation drives rapid evolution of thermal performance curves in the protist
Tetrahymena thermophila

SHORT TITLE

Rapid evolution of microbial thermal performance curves

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STATEMENT OF AUTHORSHIP:

All authors contributed to the conceptual foundation of the study. MHL, ZH, and JPG designed the study. DC contributed the fluorescent lines and expertise. MO and YY provided cell biology expertise and cell imaging. MHL, ZH, KD, KMY, and AY collected the data. MHL and JPG performed data analyses with support from DJW, RZ, and AS. JPG did the mathematical modeling and FAM the quantitative genetics. MHL and JPG wrote the first draft of the manuscript and all authors contributed significantly to revisions.

COMPETING INTERESTS

The authors declare no competing interests.

DATA AVAILABILITY

All data from this study are available on GitHub (https://github.com/JPGibert/TPC_evolution) and is archived on Zenodo (<https://doi.org/10.5281/zenodo.18602539>).

CODE AVAILABILITY

All code from this study are available on GitHub (https://github.com/JPGibert/TPC_evolution) is archived on Zenodo (<https://doi.org/10.5281/zenodo.18602539>).

ABSTRACT

Microbial respiration is a key biotic driver of climate change. Warming boosts microbial population growth, which increases biomass and respiration, potentially leading to more warming. This feedback might be disrupted by adaptation in thermal performance curves (TPCs)—whose shape describes how temperature drives growth. In this study, we uncover substantial genetic variation (G) in the intrinsic population growth rates (r) of the protist *Tetrahymena thermophila*, demonstrate a causal link between heritable variation in r and heritable variation in TPC shape, and show how this variation constrains predicted r -TPC shape evolution along specific evolutionary paths across temperatures. We also uncover Gene-by-Environment ($G \times E$) variation in r , which results in specific signatures in TPC shape and predictable temperature-dependent TPC evolution that can erode heritable variation, thus reducing future evolutionary potential. Overall, we show how temperature-dependent evolution in microbial TPC shape—a linchpin of global ecosystem function—is determined by a combination of heritable and non-heritable variation in intrinsic growth rates.

INTRODUCTION

Microbes play a central role in regulating the global carbon (C) cycle that controls climate change¹. Indeed, soil microbial respiration releases an estimated 94Pg/yr of C into the atmosphere² while microalgae fix 30-50 Pg of C/yr globally³. While these estimates come with high uncertainty², they remain five to ten times larger than the amounts of C emitted by human activity worldwide⁴, a testament to their importance in the broader global C cycle⁵. As global warming is expected to alter these microbial processes⁴, a deeper understanding of the biotic and abiotic factors influencing microbial respiration is essential in a warming world^{6,7}.

A key factor is microbial population growth, which determines total standing biomass, and consequently, total or bulk microbial respiration⁸. The thermal performance curve (TPC) of intrinsic population growth rate (r) describes how microbial population growth responds to temperature (' r -TPCs' henceforth, Fig. 1). The shape of r -TPCs is governed by temperature-dependent metabolic rates: metabolic rates increase with temperature, and so does r , until an 'optimal' temperature (T_{opt}) is reached (Fig. 1a), after which r declines as metabolic costs rise⁹ (Fig. 1a). The link between r -TPCs and metabolic rates can be extremely tight (e.g., in the genus *Tetrahymena*¹⁰), but not necessarily one-to-one –e.g., due to changes in carbon use efficiency with temperature¹¹, or differences in T_{opt} between r and metabolic rate TPCs^{12,13}. While r -TPC shape varies across species^{14,15}, they are typically unimodal¹⁶ and their shape can be described by parameters linked to thermal ecology: maximum growth rate (r_{peak}), minimum and maximum temperatures for population growth (CT_{min} and CT_{max} respectively), and rates of r -TPC increase/decrease with temperature (E_a , E_d respectively, Fig. 1b). Ultimately, these parameters reflect species-specific thermal adaptations, which influence the ability of these species to thrive in different environments¹⁶.

Understanding how TPCs evolve in novel environments is a critical challenge, as their shape determines a species' capacity to cope with environmental change. Thermal adaptation in turn depends on the evolution of intraspecific –heritable– genetic variation, and selection favoring genotypes suited to new conditions¹⁷. Quantifying heritable variation in *r*-TPC shape parameters is thus central to predicting *r*-TPC shape evolution and its effect on ecological dynamics¹⁸. To do this, we notice that the *r*-TPC represents the pattern of variation of *r* across temperatures –i.e., the reaction norm of *r*¹⁹. The question then becomes whether that reaction norm varies systematically across genotypes, and how, unlocking multiple tools from quantitative genetics.

Using quantitative genetics allows partitioning observed variation in *r* across temperatures and genotypes into (heritable) additive genetic variation (G), non-heritable environmental variation (E), and gene-by-environment interactions (GxE), if genotype-specific *r*-TPC data is available for a species. Connecting heritable variation in *r* across temperatures to heritable variation in *r*-TPC shape parameters would then further allow the estimation of heritability of *r*-TPC shape parameters, quantification of regimes of selection acting on those parameters, and predictions of shape parameter evolution under said selection. But what is the connection between heritable variation in *r* and *r*-TPC shape parameters?

In this study we address *r*-TPC shape evolution by developing and testing a conceptual model based on classic quantitative genetics of reaction norms to connect heritable *r* variation and heritable *r*-TPC variation explicitly and do so by asking whether and how E, G and GxE variation in *r* explains variation in *r*-TPC shape parameters (Fig. 1, c-e). First, environmental variation (E) in *r*, which is driven by differential expression of *r* across temperatures but not genotypes, manifests classically as completely overlapping reaction norms across genotypes

(Fig. 1c, first row). Correspondingly, genotypic r -TPCs –which are the reaction norms of r extended over multiple temperatures– should also overlap fully (Fig. 1d, first row), and present no variation in shape parameters across genotypes (Fig. 1e, first row). In other words, E variation in r alone should not lead to nor explain variation in shape parameters.

Second, genetic variation (G) in r , which arises from differential expression of r across genotypes but not temperatures, leads to additive shifts in the intercepts of each genotypic reaction norm while slopes remain equal (Fig. 1c, second row). This translates to parallel r -TPCs (Fig. 1d, second row), that show variation in CT_{min} , CT_{max} , and r_{peak} among genotypes, but no variation in T_{opt} , E_a , or E_d (Fig. 1e, second row). This inter-genotypic variation in CT_{min} , CT_{max} , and r_{peak} , is additive genetic variation, and, thus heritable. In other words, (heritable) genetic variation in r , should result in (heritable) genetic variation in CT_{min} , CT_{max} , and r_{peak} , only.

Finally, gene-by-environment ($G \times E$) interactions in r , which alters both intercept and slope of reaction norms across genotypes and temperatures (Fig. 1c, third row), result in non-parallel r -TPCs (Fig. 1d, third row), and variation in all shape parameters among genotypes (Fig. 1e, third row). Traditionally, GxE variation was considered non-heritable, but some forms of GxE have been shown to be heritable^{20–22}. So, it is possible for some fraction of this GxE variation in r , and resulting variation in r -TPCs, to also be at least partially heritable, even though it is classically assumed not to be. Overall, this conceptual model predicts that we should observe larger covariation between heritable genetic variation (G) in r and variation in CT_{min} , CT_{max} , and r_{peak} , compared to T_{opt} , E_a , or E_d . A corollary of this prediction, is that CT_{min} , CT_{max} , and r_{peak} should be more heritable than T_{opt} , E_a , or E_d . However, the latter might still show some level of heritable variation if some fraction of GxE in r is also heritable. Consequently, our model predicts that CT_{min} , CT_{max} , and r_{peak} are more likely to respond to selection than other

aspects of the TPC, which imposes specific constraints on possible r -TPC evolutionary trajectories.

To understand how r -TPC shape might evolve across temperatures we quantified E, G, and $G \times E$ variation in r in the protist species *Tetrahymena thermophila* and tested predictions from our conceptual model, revealing substantial heritable variation in both r and shape parameters, and providing insights into the structure of genetic variation and covariation in r -TPCs. Using this information, we inferred selection and potential evolutionary trajectories of r -TPC shape parameters across temperatures. Last, we identified and tested a mechanism of rapid r -TPC evolution driven by $G \times E$ variation in r , which can lead to temperature-dependent selection on r -TPC shape that may also erode heritable variation, potentially creating an evolutionary trap that constrains future microbial r -TPC evolution.

RESULTS

Heritable variation in r leads to heritable variation in r -TPC shape parameters

Intrinsic growth rates (r) from 22 genetically distinct genotypes of *T. thermophila* (Supplementary Table 1) showed strong temperature responses within genotypes ($F = 3092.70$, $D_f = 6$, Generalized η^2 (GES) effect size = 0.964, $p \leq 0.001$), significant variability across genotypes ($F = 163.65$, $p \leq 0.001$, $D_f = 21$, GES = 0.832), and significant $G \times E$ interactions ($F = 29.7$, $p \leq 0.001$, $D_f = 122$, GES = 0.840), leading to classically unimodal TPC curves that varied significantly among genotypes (Fig. 2a). Environmental variation (E) accounted for 71.7% of all observed variation in r , genetic variation (G) explained 6.2%, and Gene-by-Environment interactions ($G \times E$) explained 11.7%, with 10.5% residual variation (Fig. 2b).

These proportions are consistent with expectations for life history traits²³. After accounting for experimental inter-treatment and replicate variability, which limits the environmental and $G \times E$ effects, r was strongly heritable ($H^2_{\text{standard}}=0.76$, $H^2_{\text{cullis}}=0.95$, $H^2_{\text{piepho}}=0.91$, Supplementary Table 2, 3).

Shape parameters varied widely across genotypes. For example, genotype AXS was 3.1 times more thermally sensitive than A1868 and could grow 3.3 times faster at T_{opt} , (Fig. 2, and Supplementary Tables 4 and 5). As predicted, CT_{min} showed strong genetic covariation with r , while r_{peak} also exhibited strong covariation, although its 95% credible interval narrowly overlapped zero (Fig 3, Supplementary Table 6). In contrast there was no evidence of genetic covariation between r and either T_{opt} or E_a (Fig 3, Supplementary Table 6). Also consistent with our framework's predictions, shape parameter heritabilities were higher for CT_{min} and r_{peak} compared to T_{opt} or E_a (Fig 3). This supports the idea that heritable variation in r gives rise to heritable variation in CT_{min} and r_{peak} , while T_{opt} and E_a exhibit lower—albeit still substantial—heritability.

Consequences of G in r: selection and evolvability of r-TPC shape parameters

According to the univariate models to quantify natural selection (see Methods, Supplementary Table 7), selection varied across shape parameters and was temperature dependent: r_{peak} showed no selection at low temperatures ($<20^\circ\text{C}$, Fig. 4a, Supplementary Table 8), no directional selection to weakly positive—but no significant curvature—at intermediate temperatures (between 20 and 30°C , Fig. 4a, Supplementary Table 8), and strong positive directional selection at high temperatures (Fig. 4b, Supplementary Table 9). E_a followed a similar pattern with one difference: significant negative selection at low temperature was

observed (Fig. 4b, Supplementary Table 9). CT_{min} was under negative selection at low/intermediate temperatures but not under selection at high temperatures (Fig. 4c, Supplementary Table 10). Last, T_{opt} was not under selection at low temperatures, but showed weak then strong stabilizing selection at intermediate and high temperatures, respectively (Fig. 4d, and Supplementary Tables 10 and 11).

We found positive genetic covariances between shape parameters r_{peak} and E_a , CT_{min} and E_a , and CT_{min} and r_{peak} (Fig. 4e), so selection on highly heritable parameters CT_{min} and r_{peak} could result in correlated evolution in more weakly heritable parameters E_a , and T_{opt} . Accounting for the broader pattern of variation-covariation among shape parameters (see Methods), we confirmed differences in selection across shape parameters whose magnitude and direction shifted with temperature (Fig. 4f), leading to temperature-dependent predicted evolutionary change (Fig. 4g). Specifically, a multivariate analysis (see Methods) suggested that selection would favor higher r_{peak} at high temperatures, low CT_{min} at low temperatures, and, possibly, higher E_a at high temperatures, but no directional selection on T_{opt} (Fig. 4f), generally matching univariate predictions (Fig. 4a-d). Overall, the evolutionary responses (Δz) tracked those expected based on selection estimates (β , Fig. 4f-g), with a few exceptions that suggest indirect, correlated evolution. For example, strong positive response of E_a with temperature, despite comparatively weak selection, likely reflects correlated evolution with r_{peak} , CT_{min} or both (Fig. 4g). Also, a weak but significant expected negative response to low temperature despite weak selection likely is due to correlated evolution with CT_{min} (Fig. 4g). Despite differences in heritability, which impose constraints in r-TPC evolution, genetic correlations among shape parameters should enable temperature-dependent evolutionary responses in all of them except T_{opt} , which appears to be under stabilizing selection (Fig. 4d).

Consequences of $G \times E$ in r : sorting of standing genetic variation across temperatures

We observed significant temperature-dependent differential selection among genotypes (Fig. 5e), consistent with theoretical predictions from a simple model of genetic evolution (Fig. 5f). This confirms the hypothesis that $G \times E$ in r could drive rapid r-TPC shape evolution through selection on standing genetic variation (Fig. 5a, b). Although there were quantitative discrepancies between observed and predicted frequencies—notably at 19°C, where the model predicted a polymorphic population, but the data did not show that (Fig. 5e, f)—the model still correctly predicted qualitative changes in genetic frequencies across temperatures. These results are also robust to different experimental conditions (see Methods, Supplementary Figure 1) and fluctuating temperatures (see Methods, Supplementary Figure 2), the latter showing that temperature fluctuations could play a role in determining r-TPC evolution if r-TPCs show multiple crossing points in a short temperature span.

The experimental results also aligned with our predicted responses to selection (cf. Fig. 4 and Fig. 5). At lower temperatures, the CU4106 genotype, which has lower E_a and CT_{min} compared to AXS, became more prevalent (Fig. 5d). Consequently, the average r-TPC of the population should reflect these traits, as predicted by our multivariate model. At higher temperatures, selection favored AXS, which has higher E_a , leading to a population with an average r-TPC with higher E_a (Fig. 5d). Finally, rising temperatures initially increased genetic variance—facilitating adaptation—but then reduced it as the CU4106 genotype became less prevalent (Fig. 5e, f), potentially constraining future adaptive evolution.

DISCUSSION

Our study reveals considerable heritable genetic variation in *Tetrahymena thermophila*'s intrinsic population growth rates and several *r*-TPC shape parameters (Fig. 2, 3), thus enabling clear potential for thermal adaptation to new climates (Fig. 4). The observed *intraspecific* variation in *r*-TPC shape parameters sometimes rivals reported *interspecific* variation across multiple protist species¹⁵. We also found that most *r*-TPC shape parameters are under different selection regimes across temperatures (Fig. 4), with CT_{min} and r_{peak} being more likely to respond to selection than E_a and T_{opt} . However, less heritable parameters can evolve through correlated evolution (Fig. 4). Specifically, colder temperatures select for lower CT_{min} , r_{peak} , and E_a , and warmer temperatures favor high r_{peak} and E_a (Fig. 4). Finally, we demonstrated that $G \times E$ in *r* can drive rapid, predictable shifts in population genetic makeup across temperatures, facilitating *r*-TPC evolution (Fig. 5), consistent with a “plasticity drives adaptation” scenario²⁴.

While the evolution of microbial TPCs is likely shaped by local adaptation²⁵, how *r*-TPCs will adapt to rising temperatures remains uncertain. Several evolutionary paths have been proposed: 1) ‘Colder-is-Better’ (CIB), where warming reduces population growth, lowering r_{peak} , T_{opt} , and E_a ²⁶; 2) ‘Warmer-Is-Better’ (WIB), where higher temperatures favor increased r_{peak} and T_{opt} ²⁷; and, 3) Generalist-Specialist-Tradeoff (GST), where species either evolve towards rapid growth within a narrow temperature range –i.e., temperature specialists with higher r_{peak} , higher CT_{min} , and lower CT_{max} – or slower growth over a broader range –i.e., temperature generalists with lower r_{peak} , lower CT_{min} , and higher CT_{max} ²⁸. Tests of these hypotheses often rely on interspecies comparisons^{10,25}, which would be technically capable of accounting for intraspecific variation within shape parameters and genetic associations that ultimately determine evolutionary trajectories within clades, if such data was readily available. Our study provides that perspective and precisely the type of data that has been missing, and shows that by

accounting for genetic covariances, E_a evolution is likely driven by correlated evolution in response to positive selection on r_{peak} (Fig. 4f,g), despite no selection acting on T_{opt} (Fig. 4d,f,g), thus contradicting WIB. Positive selection in r_{peak} (Fig. 4a,b) and the absence of positive selection in CT_{min} at higher temperatures (Fig. 4c), in turn contradict CIB and GST. These results therefore suggest that neither WIB, CIB or GST, can explain predicted evolutionary responses in *T. thermophila*, emphasizing the importance of intraspecific variation and genetic associations to fully understand r -TPC evolution.

When $G \times E$ in r is prevalent, we showed that thermal adaptation can occur rapidly through temperature-dependent selection on r -TPC genotypes (Fig. 5), resulting in reduced genetic variation, which could slow down, or impede, future adaptation²⁹. In other words, we show that phenotypic plasticity in r could lead to adaptive evolution in r -TPCs in novel climates, but at the expense of genetic diversity. That this form of r -TPC evolution could be anticipated from simple models of genetic evolution, is striking, considering how little information beyond the r -TPC of each genotype the model accounts for. Yet, r -TPC adaptation in deep time has been suggested to occur gradually across six different *Tetrahymena* species¹⁰, directly countering our claim that r -TPCs could evolve rapidly through temperature-dependent selection. We argue that this form of rapid r -TPC evolution may only play out during fast-changing environmental conditions, not always result in long-term r -TPC change, or be masked by reversals in selection. In fact, selection often “erases its traces”³⁰, and is strongest under novel conditions, weakening as populations adapt³¹—i.e., “the paradox of stasis”³⁰.

The temperature-dependent loss of genetic diversity in adaptive r -TPC evolution could influence the survival of genetically depauperate species in a warming environment. However, whether this is the case in nature depends on quantifying the processes that generate diversity

(mutation, gene flow, recombination) in r and r -TPC shape parameters in the wild, as well as evaluating when these processes might counter the loss of genetic diversity following episodes of thermally induced r -TPC evolution. *T. thermophila* has the lowest reported base substitution mutation rate of all species³² and limited gene flow³³, but even weak dispersal or rare mutation events could contribute to r and r -TPC variation and rescue genetic diversity in very large, rapidly reproducing populations. The prevalence of sexual reproduction in nature is debated –e.g. ~50% of wild individual *T. thermophila* cannot reproduce sexually and some *Tetrahymena* lineages have lost this ability entirely³⁴ – but likely important if present. In fact, *T. thermophila* possesses two separate genomes, a germinal line (in micronucleus) and an expressed genome (in macronucleus). Our study focuses on macronuclear variation, but asexual reproduction following sex results in random distribution of macronuclear alleles to daughter cells—i.e. phenotypic assortment—potentially generating novel variation in r and r -TPCs³⁵. Moreover, seemingly lost genetic variation could persist in the population in other individual's micronuclei and reemerge following sexual reproduction³⁶. Fully understanding thermal adaptation in nature thus requires exploring these processes' interplay, yet, for most organisms, they remain poorly understood.

We quantified TPCs without acclimation to each temperature before conducting growth assays. By definition, these TPCs must characterize an *acute* response. Because acclimated TPCs can differ from acute ones^{37,38}, whether our TPCs meaningfully quantify change in performance across temperatures compared to fully acclimated TPCs needs to be addressed. Thermal acclimation occurs within a single generation in *Clycidium glaucoma* with regard to cell size dynamics³⁹, which has been shown to influence population growth^{40,41}. Knapp et al.⁴² further show rapid acclimation in less than 2 generations in the bacterium *Escherichia coli*. In *T. thermophila*, generation time varies from 2 to 5hrs, and significant population growth occurs 6-

24hr –but more commonly, 12-24hr– post seeding, and rarely before then (lag period)⁴³. It is therefore possible that our TPCs show a pattern of growth for near-acclimated populations, but that cannot be directly determined by our data. However, shape parameter values from acclimated TPCs for *T. thermophila* closely match our own, despite differences in experimental contexts, media, and procedures, which provides independent validation for our TPCs.

There is additional evidence suggesting that these TPCs meaningfully reflect performance, at least in protists. Wieczynski et al.¹⁵ used a very similar assay protocol to quantify the (acute) TPCs of 14 ciliate species –albeit with variation in the number of days elapsed between quantifying initial and final densities– then showed that their shape could be predicted by species traits (e.g., body size, shape, optical properties). In turn, these TPCs accurately predicted temperature-driven shifts in community dynamics, diversity, trait distributions, and ecosystem function. Moreover, Vasseur et al.⁴⁴ showed that a TPC quantified using the same procedure as in Wieczynski et al.¹⁵ –the same used in the present study– also accurately predicts population growth, persistence, and extinction risk in *Paramecium caudatum* across temperatures. Together, these results suggest that our TPCs might be capturing more than the acute stress responses of unacclimated organisms, and clearly reflect ecologically relevant performance across temperatures, even though at temperatures much higher than 22°C the individuals might be experiencing heightened thermal stress. This heightened thermal stress may ultimately impair their ability to survive and reproduce at rates higher than would be expected in a natural setting, which is a limitation of the present study.

While bodies of water in *T. thermophila*'s native range are warming significantly⁴⁵ and experiencing more frequent and extreme heatwaves⁴⁶, water temperature typically remains 2-5°C cooler than air temperature⁴⁷. Thus, some of the warmer temperatures in our study may be

uncommon in *T. thermophila*'s native range. However, strong temperature effects observed at low temperatures suggest these patterns can still operate in many *T. thermophila* populations. Also, previous research showed that variation in *r*-TPC shape across species could be collapsed into a single dimension of variation⁴⁸, suggesting that some of the covariation observed in this study between shape parameters could be spurious. However, if the *r*-TPCs collected here could in fact be collapsed into a single axis of variation, we would have expected strong and multiple covariations between most if not all shape parameters—which was not the case (Fig. 4e, f). Interestingly, most of the genetic variation in shape parameters was observed, in CT_{min} and r_{peak} , but these two parameters were at best weakly correlated to one another. So, perhaps intraspecific *r*-TPC variation buck the trend observed across species⁴⁸, but more research is needed to clarify this point.

Overall, TPCs control the fate of populations⁴⁹, ecological interactions⁵⁰, food web dynamics^{41,51}, and ecosystem processes⁵². Yet, TPC evolution in a rapidly warming world remains conspicuously unknown. Here, we show that *r* intraspecific variation can drive temperature-dependent evolution in microbial *r*-TPCs and have consequences for rapid shifts in population genetic makeup. Our study emphasizes the importance of temperature in mediating rapid microbial evolutionary change as we grapple with understanding and predicting organismal responses to an increasingly warmer world.

METHODS

Study system and genotypes

Tetrahymena thermophila is a freshwater ciliate protist found across the Northeastern United States³³ and part of a cosmopolitan genus⁵³. Protists are unicellular Eukaryotes that

dominate oceanic biomass and rank third in terrestrial biomass, collectively comprising twice the biomass of the Animal Kingdom⁵⁴ and are key contributors to global ecosystem functioning⁵⁵. While no single species is representative of the entire group, studying thermal adaptation in *T. thermophila* might provide valuable insights into this important lineage.

We sourced 22 unique *T. thermophila* genotypes: 19 from the Washington University *Tetrahymena* Stock Center (<https://sites.wustl.edu/tetrahymena/>) and 3 from the Chalker lab at Washington University (Supplementary Table 1). These genotypes vary in geographic origin, exhibit well-documented genetic differences (Supplementary Table 1), and were selected to capture existing genetic variation rather than for their functional significance. While systematic biases in genetic distances between genotypes could influence analyses, to our knowledge, there is no genome-wide molecular divergence data for all genotypes used. To minimize the chance of inserting this form of bias, we aimed for a broad representation of genetic backgrounds, strains capable of both sexual and asexual reproduction, and both laboratory-generated and natural genotypes (Supplementary Table 1). Despite these efforts, most genotypes are artificial mutants, so our assemblage likely contains less variation than natural populations –which show significant amounts of genetic divergence even between neighboring populations³³. Thus, our findings likely are only a conservative estimate of potential thermal adaptation in nature.

Quantifying r , r -TPCs, and TPC shape parameters

The Washington University *Tetrahymena* Stock Center maintains cultures frozen and thaws them up before sending. Upon reception, we transferred the genotypes from axenic Proteose Peptone growth medium to Timothy Hay growth medium inoculated with a bacterial community from Duke Forest Gate 9 pond/Wilbur pond (Lat 36.013914, Long -78.979720) –

described elsewhere⁷— and a wheat kernel, as a Carbon source⁵⁶. Cultures were allowed to grow to carrying capacity (~4 days) in Percival AL-22 growth chambers in a 12hr day-night cycle at 22°C, prior to quantifying their TPCs. We quantified TPCs by conducting growth assays at seven temperatures (13, 19, 22, 25, 30, 32, 38°C) in 3 cm diameter Petri dish microcosms containing 3 mL of growth medium. Cultures were transferred directly to their assay temperatures without an acclimation period. Each genotype-temperature combination was replicated six times, totaling 924 microcosms. These experimental temperatures span below and above the average growing season temperature in *T. thermophila*'s native range (~23°C)⁵⁷. Microcosms were initialized at densities of 1 ind/mL in 3mL Petri dishes, which results in a concentration that supports exponential growth for 1-2 days^{40,41,43}. After 24 hours, we conducted whole-population counts using a stereomicroscope (Leica M205C) and calculated r across temperatures as $\log(\text{final density}/\text{initial density})/\text{time}$, with time = 1 day¹⁵.

To estimate r -TPC shape parameters, we fitted a modified Sharpe-Schoolfield model using the “nls.multstart” v1.3.0 package in R⁵⁸. We did so by converting the classic exponential form of the Sharpe-Schoolfield model to its log form, then fitted with an intercept, a :

$$\log(r) = a + \frac{E_a}{k_{boltz}} \left(\frac{1}{T_{ref}} - \frac{1}{T} \right) - \log \left(1 + e^{\frac{E_d}{k_{boltz}} \left(\frac{1}{T_d} - \frac{1}{T} \right)} \right),$$

where k_{boltz} is the Boltzmann constant ($=8.6 \times 10^{-5} \text{ eV/K}$), T_{ref} is a temperature of reference (298.15°K), and parameter T_d represents the temperature at which the performance is halfway down past T_{opt} relative to its peak. We dealt with negative r values by adding a small positive scale value prior to log-transforming. The magnitude of the scale can alter the estimation of the TPC shape, so we tried different scale values to test the robustness of our results, which did not qualitatively change under different scales (Supplementary Table 1 and Supplementary Figures

3-5). Shape parameters r_{peak} , CT_{min} and T_{opt} were estimated numerically using the predicted TPC fit.

Having focused on a range of temperatures that is most likely to be experienced by the species in their native range, our data lacks good representation of extremely warm temperatures ($T > 38^{\circ}\text{C}$) to confidently estimate parameters E_d and CT_{max} . We therefore focused on parameters E_a , r_{peak} , CT_{min} and T_{opt} for all analyses (Fig. 1b). These parameters govern the rising portion of the r -TPC (Fig. 1b, green) –i.e., the “operational temperature range”¹³. For genotype B2192 III, T_{opt} could not be estimated without extrapolation and was considered, and was therefore not calculated. Some analyses cannot be conducted with NAs in the data, so this specific genotype was excluded from most analyses. However, we did not find a significant effect of removing that genotype from analyses whenever we could include the genotype.

Link between heritable genetic variation in r , and r -TPC shape parameters

We quantified E, G and $G \times E$ variation in r using function *gxeVarComps()* in R package *statgenGxE* v1.0.5. on the observed r values for all genotypes across temperatures. The function fits a series of mixed effects models with r as the response variable, and temperature, genotype, and their interaction as fixed terms to calculate effect sizes and statistical significance, and then refits the model with those terms as random effects to estimate variance components (E, G, and $G \times E$). The fraction of heritable genetic variation in r is classically defined as broad-sense heritability, $H^2 = G / (E + G + G \times E)$. However, this expression does not account for inter-treatment and replicate variability, which inflates E relative to G, and underestimates H^2 ^[59,60]. Other standard formulations for H^2 exist that address these issues (Supplementary Table 2) and were calculated using package *statgenGxE*.

Next, we tested the predictions of our verbal model (Figure 1) linking heritable variation (G) in r , to heritable variation in r -TPC shape parameters. Since each r -TPC yields a single set of shape parameters, only G variation in these parameters can be quantified. All other components of variation (e.g., E, $G \times E$, residual variation and error) were thus treated as residual variance. We notice that this approach is conservative: it treats all unexplained variation (including any heritable GxE that may exist) as non-genetic, and hence, as part of the denominator, which underestimates heritability. To evaluate the model's prediction that G variation in r results in greater G variation in r_{peak} and CT_{min} compared to E_a , and T_{opt} (Fig. 1e), we tested whether r_{peak} and CT_{min} exhibited stronger covariance with r than did E_a , and T_{opt} using a multivariate Bayesian Generalized Linear Mixed Model. The model includes mean-scaled values of r , E_a , r_{peak} , CT_{min} and T_{opt} , as jointly modeled response variables, each with their own fixed mean, and Genotype as a random effect, which allowed us to estimate variances and covariances. The model was implemented in the R package MCMCglmm v.2.36⁶¹. Variance-covariance priors were obtained by bootstrapping r -TPC data 100 times within each genotype and re-fitting the Sharpe-Schoolfield model to each bootstrap replicate, generating a population of shape parameters for each genotype (Supplementary Figure 6). Using non-informative inverse Wishart priors⁶² did not alter our results. Finally, if G variation in r results in greater G in r_{peak} and CT_{min} , these shape parameters also should exhibit greater heritability than E_a , and T_{opt} . To test this, we calculated the broad-sense heritability in r -TPC parameters using the standard expression $H^2 = G / (G + \text{residual variance})$ using 1000 posterior samples of G and the residual variance matrix from the MCMCglmm model.

Consequences of heritable variation: selection and evolutionary potential of r -TPC parameters

To understand how r -TPC shape parameters can evolve under selection, we quantified 1) the direction, form, and magnitude of selection on all four shape parameters, 2) whether and how temperature influenced selection, and 3) their potential evolutionary responses across temperatures, to said selection. We used two approaches: one that allows for the estimation of non-linear selection (e.g., quadratic, stabilizing selection), but neglects genetic correlations between parameters and thus cannot make predictions about possible evolutionary trajectories, and another that accounts for genetic correlations, does not account for non-linear selection, but enables predictions of evolutionary responses across temperatures. The combination of both gives us the most complete picture possible.

In the first approach, we quantified selection by characterizing the adaptive landscape as the relationship between shape parameter values and absolute fitness, assumed here to be a function of r ^{63,64}. This assumption is justified because r reflects average birth and death rates, two major fitness components^{65,66}. We modeled this relationship for all shape parameters across genotypes using polynomial regression with r as the response. We obtained linear and quadratic effects (i.e., non-linear selection) for each mean-standardized shape parameter, as well as additive and interactive temperature effects (Supplementary Table 3). Quadratic regression coefficients were multiplied by two⁶⁷. A positive relationship between r and a shape parameter indicates positive directional selection, a negative relationship indicates negative directional selection, and no relationship suggests no directional selection. Stabilizing selection produces a concave-down relationship, while disruptive selection manifests as concave-up, where extreme values have higher fitness⁶⁵.

The second approach predicts r -TPC shape evolution across temperatures while accounting for genetic correlations among shape parameters. It uses the multivariate breeder's⁶⁴

and Price's⁶⁸ equations to estimate direct (i.e., acting on the focal parameter) and indirect selection (i.e., acting on genetically linked parameters⁶⁹). This is achieved by estimating the response to selection of all shape parameters, $\Delta\mathbf{z}$, as their covariance with fitness, i.e., the Price equation – and back-estimating selection gradients, $\boldsymbol{\beta}$, via the multivariate breeder's equation.

Operationally, this approach requires estimating a variance-covariance matrix $\mathbf{G}_{\mathbf{zw}}$, which includes the genetic variance-covariance matrix ($\mathbf{G}$) of the focal traits (r -TPC shape parameters) as its first four rows/columns, and their genetic covariation with fitness (W) as the last row/column^{63,64,70}. We conducted the analysis using either r as the measure of fitness, or the finite rate of increase, $R=\exp(r)$, the latter providing numerical stability in and around the points where r goes from negative to positive (both metrics led to similar results). From the Price equation, the last row/column of $\mathbf{G}_{\mathbf{zw}}$ equals the vector of predicted trait change, $\Delta\mathbf{z}$. To estimate $\boldsymbol{\beta}$, we used the $\mathbf{G}$ and $\Delta\mathbf{z}$ components of $\mathbf{G}_{\mathbf{zw}}$ in the multivariate Breeder's equation, $\Delta\mathbf{z} = \mathbf{G}\boldsymbol{\beta}$, where $\boldsymbol{\beta}$ is the selection gradient⁶⁴, which can be rearranged as $\boldsymbol{\beta} = \mathbf{G}^{-1} \Delta\mathbf{z}$, to estimate selection.

Positive/negative Δz_i values indicate an increase/decrease in the i -th shape parameter, while positive/negative β_i , indicates positive/negative selection operating on the i -th shape parameter. Alignment between Δz_i and β_i indicates direct responses to selection⁷¹, while misalignment suggest the effect of selective tradeoffs with other correlated traits.

To estimate $\mathbf{G}_{\mathbf{zw}}$ for each temperature, we used subsets of r per genotype for each temperature. While shape parameters do not change across temperatures, r does, so the $\mathbf{G}_{\mathbf{zw}}$ matrices only differed in their last row/column across temperatures, i.e., the covariances between shape parameters and fitness ($\Delta\mathbf{z}$). For interpretability, $\mathbf{G}_{\mathbf{zw}}$ was calculated as $\mathbf{G}_{\mathbf{zw}} = \mathbf{S}\mathbf{L}\mathbf{S}^*(1/2F)$ where $\mathbf{L}$ is a between-genotype covariance matrix, $\mathbf{S}$ is a diagonal matrix containing the inverse

of each trait's standard deviation, and F is the inbreeding coefficient, which equals 1 for clonal lineages⁷². This scaling ensures that the resulting β and Δz will be comparable across traits⁷¹.

Uncertainty in $\mathbf{G}_{zw}$ entries was quantified using a Bayesian posterior distribution of $\mathbf{L}$ matrices (and hence $\mathbf{G}_{zw}$). We used a multivariate normal likelihood function and non-informative inverse regularized Wishart priors ($\text{nu}=\text{traits}+1$)⁶², implemented in *evolqg* v3.0 R package⁷³. Using *MCMCglmm* instead of *evolqg* did not affect our results (Supplementary Figure 7). We took 1000 posterior samples for $\mathbf{G}_{zw}$ to calculate Δz , $\mathbf{G}$, and β , and 95% maximum density intervals for all estimates.

Consequences of $G \times E$ in r for rapid r -TPC evolution

Because $G \times E$ variation in r is predicted to result in non-parallel r -TPCs (Fig. 1), slight differences in intrinsic growth rates across temperatures can drive temperature-mediated differential selection among genotypes (Fig. 4a, b). Treating r as proxy for absolute fitness⁶⁴, this selection should be strongest at temperatures where r -TPCs differ the most, i.e., the temperatures with the largest differences in relative fitness.

We empirically tested for temperature-dependent selection among genotypes by setting up an experimental evolution assay with two fluorescently marked strains (AXS and CU4106, Fig. 4c) with different r -TPCs and relative fitness (Fig. 4d, Fig. 4d inset) across six temperatures (19, 22, 25, 30, 32, 38°C). Each temperature treatment was replicated seven times, and with two additional single-strain controls per temperature. The genotypes cannot mate because they belong to the same mating type (mating type VII, Supplementary Table 1), so only clonal reproduction was possible. We initialized our microcosms at equal densities (5 ind/mL in mixture treatments, 10 ind/mL for single-genotype controls). After 48 hours, we added Cadmium

Chloride to induce fluorescence⁷⁴, confirmed on a Leica Thunder Cell Culture inverted microscope equipped with an HC PL APO 63X/1.40 N.A. oil-immersion objective lens (Fig. 4c and Supplementary Figure 8). Thirty minutes later, we cytometrically censused each sample (Novocyte 2000R). While fluorescently tagged genotypes can lose their ability to fluoresce over time, they carry a paromomycin resistance gene, allowing selection through paromomycin exposure (100µg/mL) prior to counting and estimating relative frequencies based on fluorescence (Supplementary Figure 9 and Supplementary Table 12). Antibiotics can negatively affect protists and the bacterial communities they feed on, so we replicated this experiment in paromomycin-free conditions, which did not qualitatively alter our results (Supplementary Figure 1).

Finally, observed changes in genetic frequencies were compared to predicted frequencies by a classic model of genetic evolution in discrete time parameterized with the r -TPC of both experimental genotypes (Fig. 4f). The model tracks the frequency of each strain f_i , and assumes that their absolute fitness, W_i , is a function of their average intrinsic growth rate $r^{63,64}$. The frequency of each strain in the population is determined by the classic recursive equation, $f_i(t + 1) = f_i(t)(W_i/\bar{W})$, where $\bar{W}$ is the average fitness of the population ($\sum_{i=1}^n f_i W_i = \sum_{i=1}^n f_i r_i$), and $W_i/\bar{W}$ is the relative fitness of i -th genotype. We used each genotype's r -TPC (AXS, CU4106, Fig. 4d) to estimate $W_i/\bar{W}$ across temperatures (i.e., replacing r_i). Because r -TPCs are interpolated, this was done for an arbitrarily large number of temperatures in the temperature range for which data was available, but in the main text we present the predictions of the model only for those temperatures at which the experiment was conducted. This model was then forward solved under two scenarios: one with constant temperature and another with randomly fluctuating temperatures drawn from a normal distribution with known mean and variance to assess whether predictions based on constant-temperature r -TPCs hold under more realistic

thermally variable conditions. This minimal model assumes that clonal sorting is primarily driven by differences in r-TPCs, disregarding density-dependent processes (e.g., explicit competition between genotypes), frequency-dependent selection, sexual reproduction, mutation, and most ecological processes. All data that supports this study is available with the identifier: <https://zenodo.org/records/18602540>⁷⁵. The code to generate the figures for this study is also available with the identifier: <https://zenodo.org/records/18602540>⁷⁵.

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FIGURE CAPTIONS

Figure 1. Conceptual model of how heritable and non-heritable variation in r explains variation in r -TPC shape parameters.

(a) General shape of the r -TPC. (b) r -TPC shape parameters. Blue-colored shape parameters, measured in this study, include r_{peak} , E_a , CT_{min} and T_{opt} . (c) environmental variation (E) in r comes from expressing different r across temperatures but not genotypes, and leads to overlapping reaction norms (E), genetic variation (G) in r results from expressing different r across genotypes, but not temperatures, and leads to additive shifts in reaction norm intercepts, but not slopes (E+G), gene-by-environment interactions ($G \times E$) results from patterns expression of expression of r changing across temperature, leading to shifts in slopes of reaction norms (E+G+ $G \times E$). (d) Because- r TPCs are multi-temperature reaction norms of r , classic quantitative genetics can explain how E, G, $G \times E$ in r influence r -TPC shape: E in r produces fully overlapping r -TPCs, G in r result in non-overlapping r -TPCs with genotype-specific intercepts but not slopes, and $G \times E$ leads to r -TPCs that vary in intercepts and slopes. (e) These r -TPC shape signatures create variation in r -TPC shape parameters: overlapping r -TPCs show no variation in shape parameters, additive shifts in r -TPC intercepts lead to additive (heritable) variation in r_{peak} , CT_{min} , and CT_{max} , while slope and intercept variation in r -TPCs creates variation in all shape parameters.

Figure 2. Observed r -TPC variation across genotypes: (a) Observed r -TPCs for all 22 genotypes. Dots represent observed r values, bold lines represent Sharpe-Schoolfield model fits,

and dash lines represent average TPCs across all experimental genotypes. (b) All 22 TPCs are superimposed and dashed lines represent the average r-TPC. Inset: amount of variation due to residual, $G \times E$, G , and E variation in r .

Figure 3. Covariation between r and r-TPC shape parameters and parameter heritability.

(a) Posterior distribution of genetic covariances between r and each r-TPC shape parameter. Bars represent 95% credible intervals. (b) Posterior heritability distribution for all four r-TPC shape parameters.

Figure 4. Quantifying selection on r-TPC shape across temperatures.

(a) Estimated adaptive landscape across temperatures (i.e., change in fitness with a change in the underlying shape parameter) for r_{peak} . Color indicates temperature bins (for illustration purposes, blue: 10—20°C, yellow: 20—30°C, red: 30—40°C) (b) As in a, but for E_a . (c) As in a, but for CT_{min} . (d) As in a, but for T_{opt} . (e) Observed genetic associations between shape parameters. Each dot is a genotype color-coded as in Fig. 2. In gray, 95% confidence ellipses. ρ represents correlation coefficients. (f) Estimated multivariate selection coefficient (β , 95% maximum density intervals) for all shape parameters across temperatures, or eco-evo landscapes (MacColl 2011). (g) Predicted evolutionary change (Δz , 95% maximum density intervals) for all shape parameters, across temperatures.

Figure 5. Rapid evolution driven by $G \times E$ in r-TPC shape and differential selection on standing variation across temperatures.

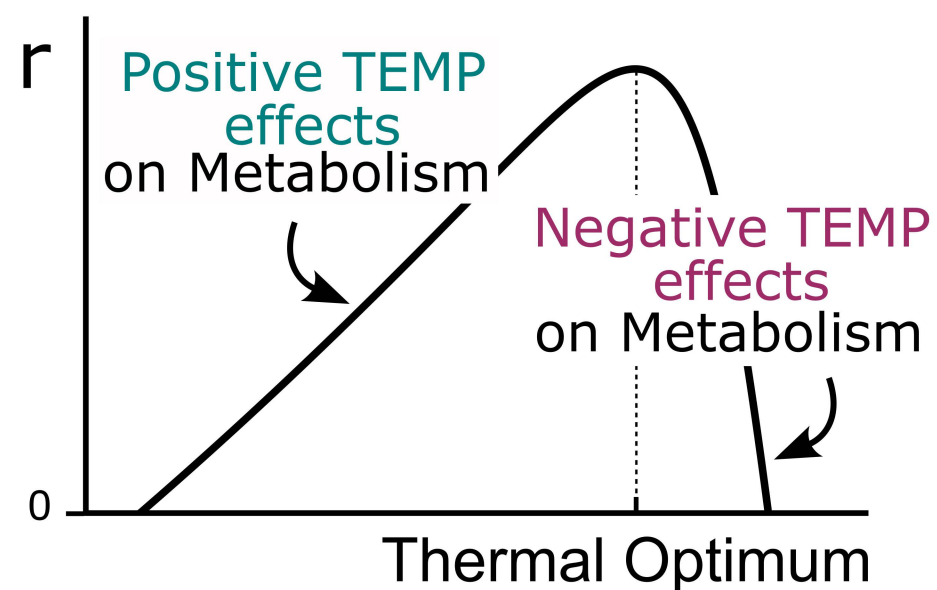
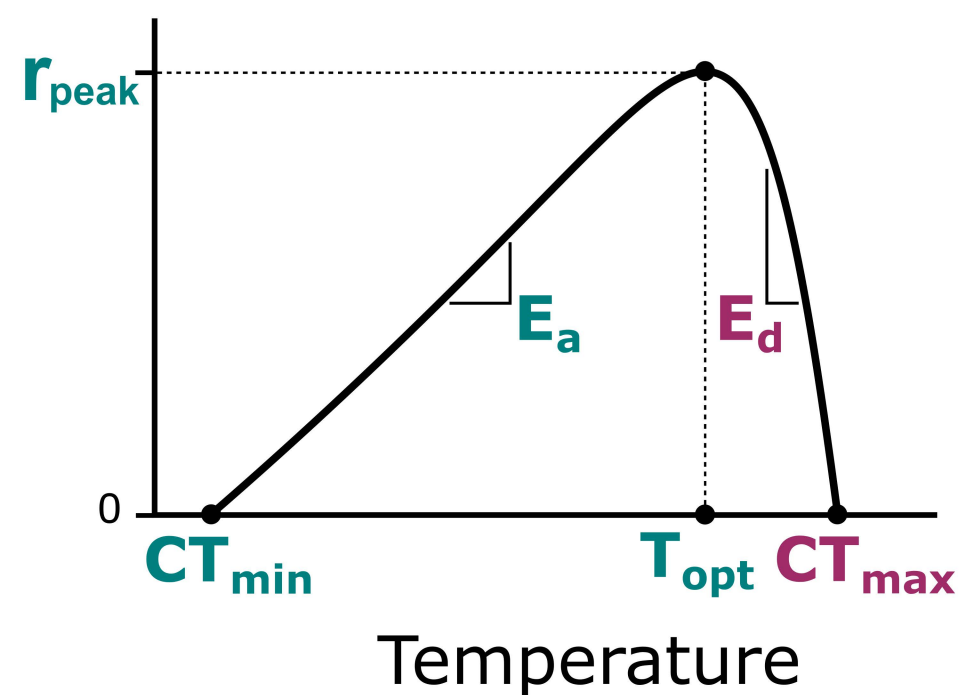
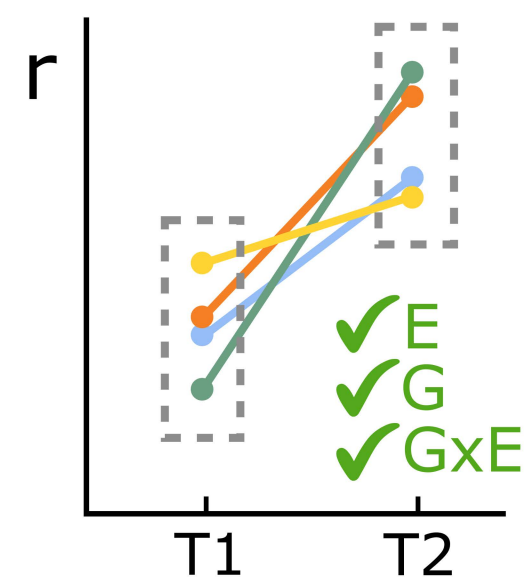
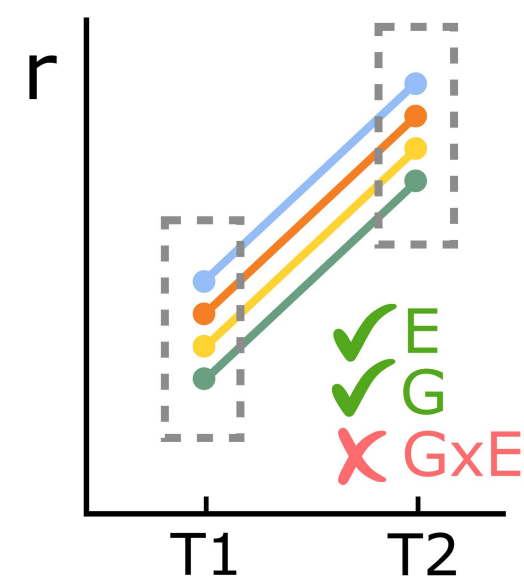
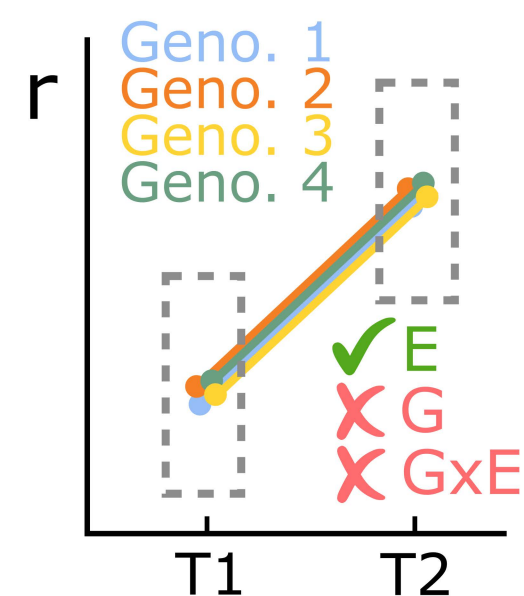
(a) $G \times E$ variation in r-TPCs leads to differential growth of each genotype across temperatures. (b) Differential growth across temperatures leads to differential selection across environments and rapid shifts in genetic frequencies across temperatures. (c) Differential Interference Contrast (DIC) microscopy for two genotypes of the protist *Tetrahymena thermophila* (CU4106 and AXS) with fluorescence microscopy image overlaid. Only AXS fluoresces (green) due to the expression of Yellow Fluorescent Protein (YFP) while both strains autofluoresce (A.Flu, in pink). White bar represents 10 μm . (d) r-TPC for genotypes CU4106 and AXS. Inset: Measures of relative fitness for both CU4106 and AXS. This predicts an increase in AXS frequency relative to CU4106 at intermediate temperatures relative to low or high temperatures. (e) Observed genetic frequencies across temperatures. (f) Predicted genetic frequencies across temperatures (black bar) and predicted frequencies corrected for genotype loss at 19°C (colored bars). The model qualitatively captures the observed pattern of genotypic change across temperatures but overestimates the persistence of genotype AXS at 19°C temperature. We used that to tighten the model predictions presented in the color bars.

Editor Summary

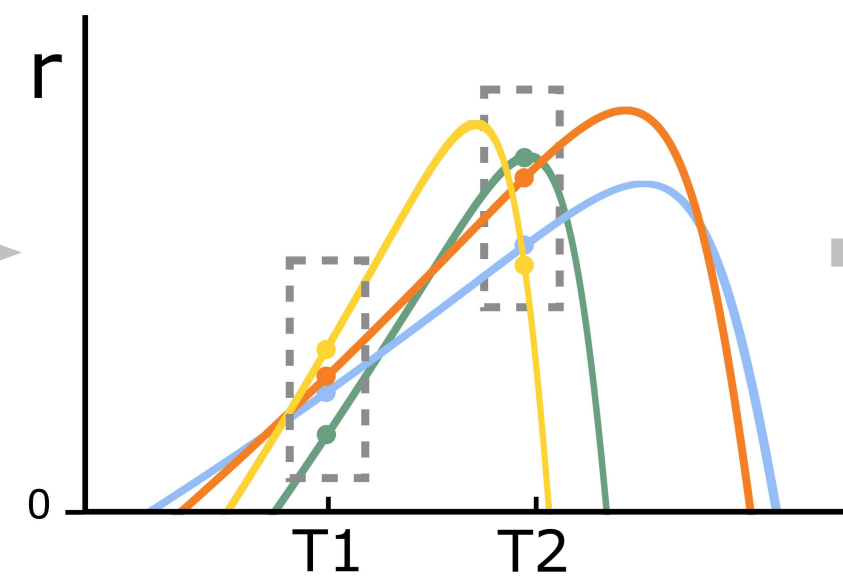
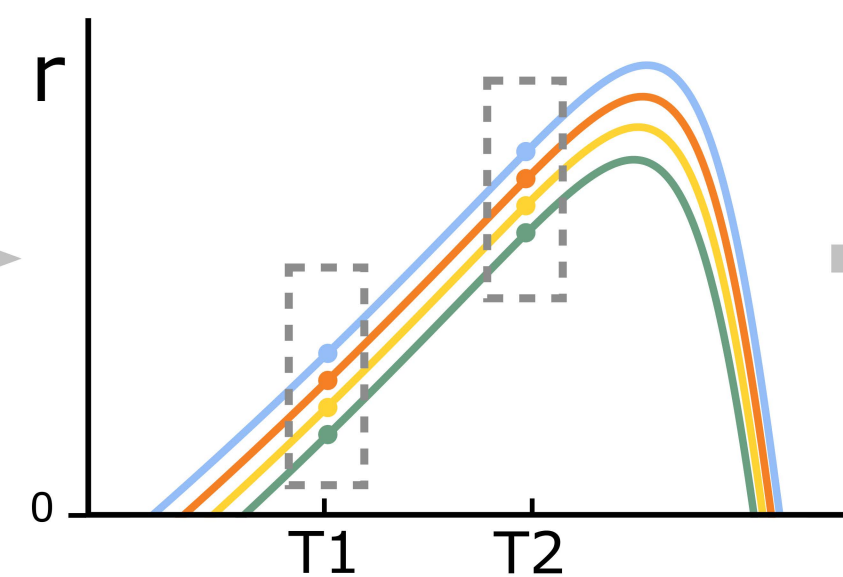
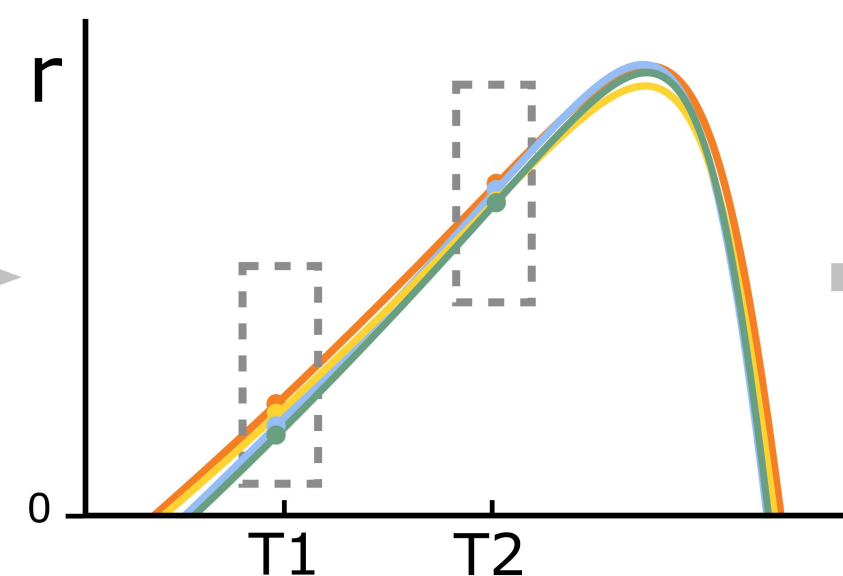
Quantification of heritable and non-heritable variation in intrinsic growth rates reveals temperature-dependent adaptive evolution of microbial thermal performance curves, and gene-by-environment constraints on thermal adaptation in a protist system.

Peer Review Information

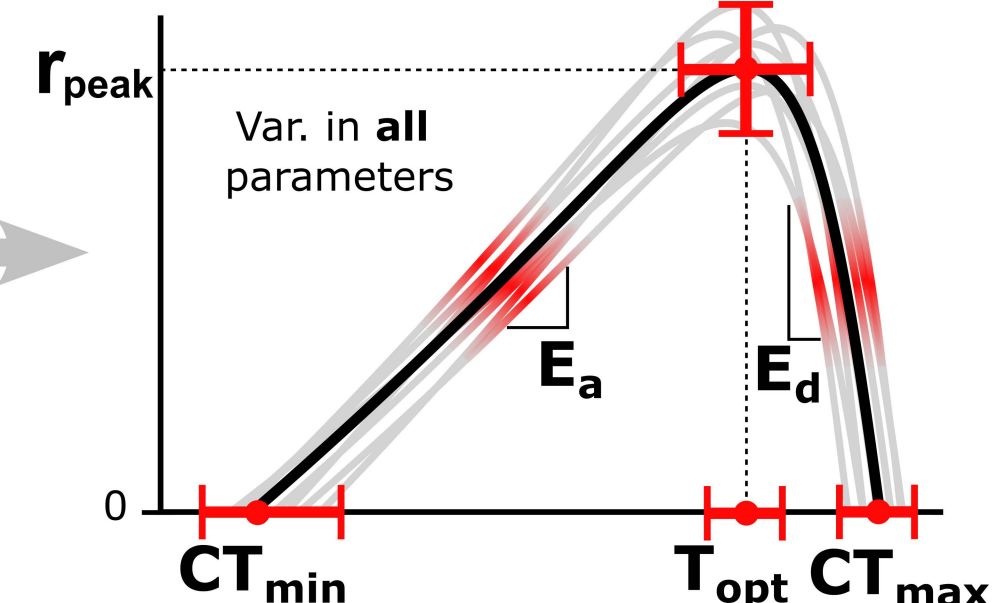
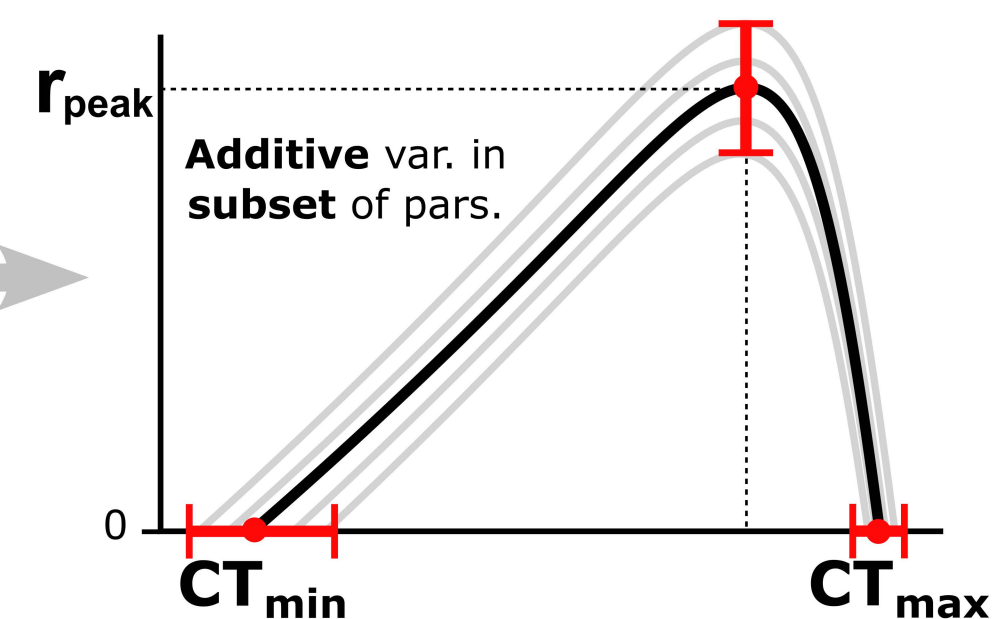
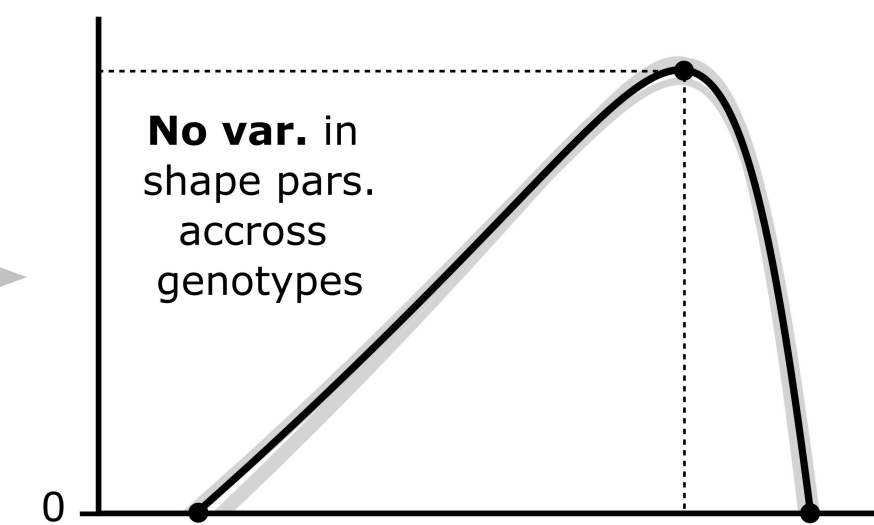
Communications Biology thanks Daniel Padfield and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. Primary Handling Editor: Michele Repetto. A peer review file is available.

a. General TPC shape**b.** TPC shape parameters**c.** Intraspecific variation in r 

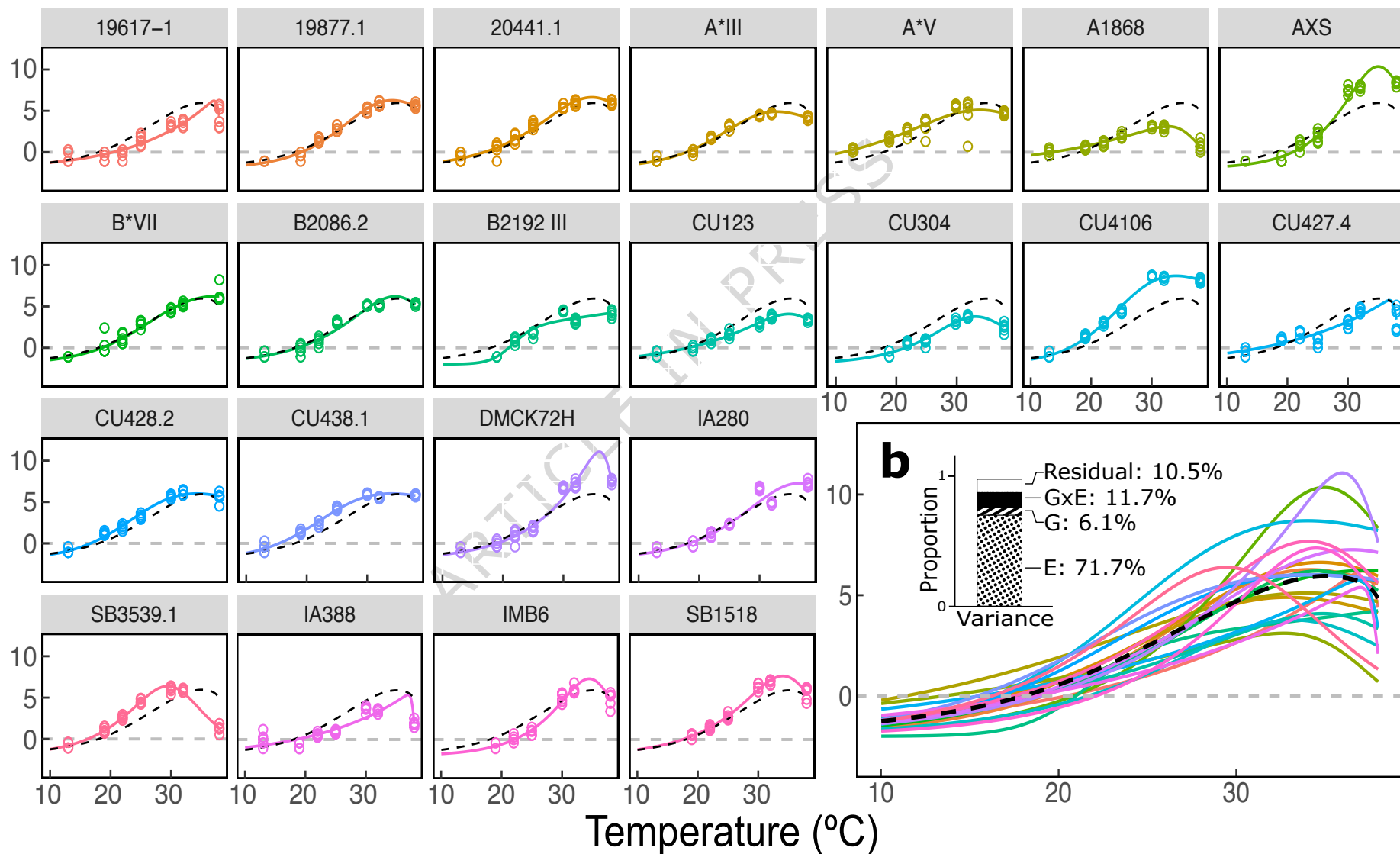
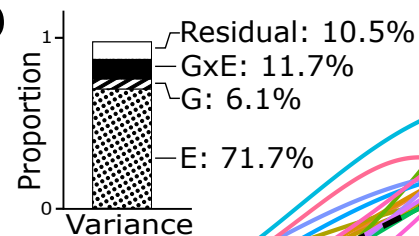
Temperature

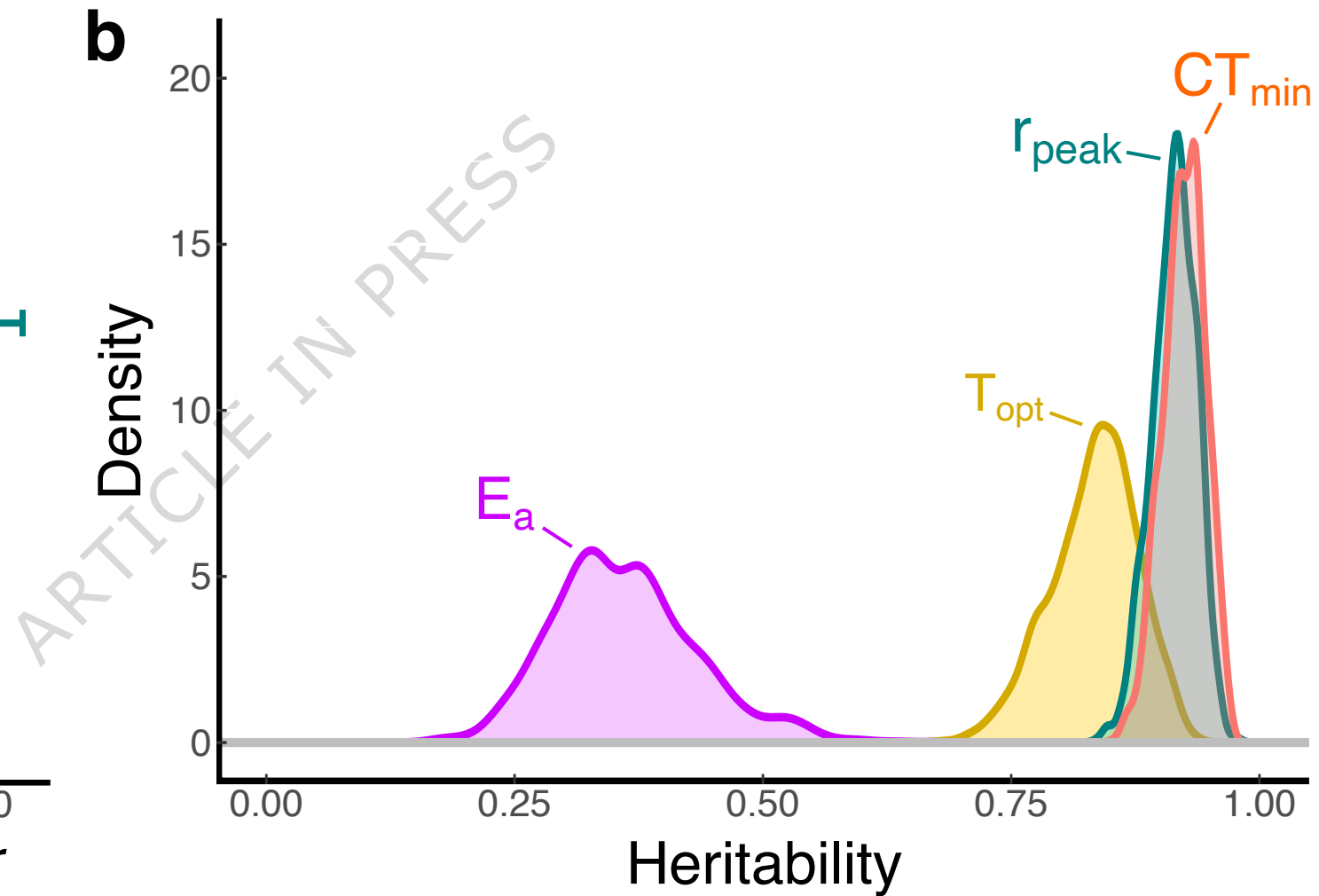
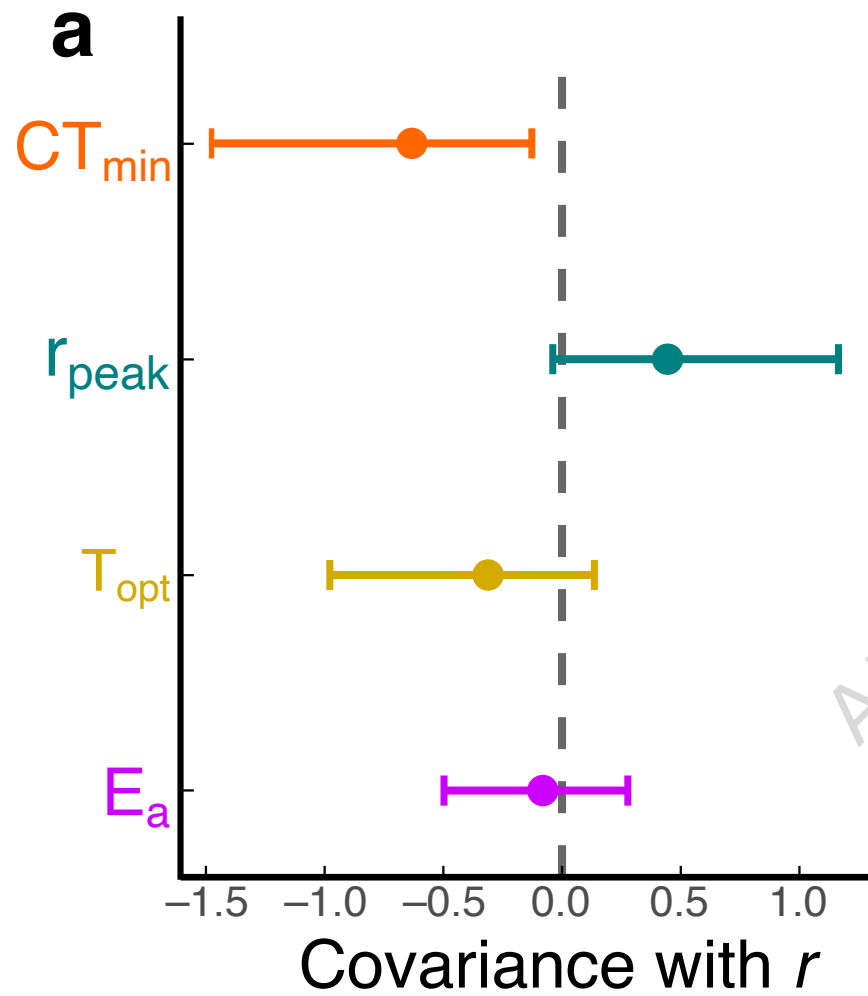
d. Associated r -TPC shapes

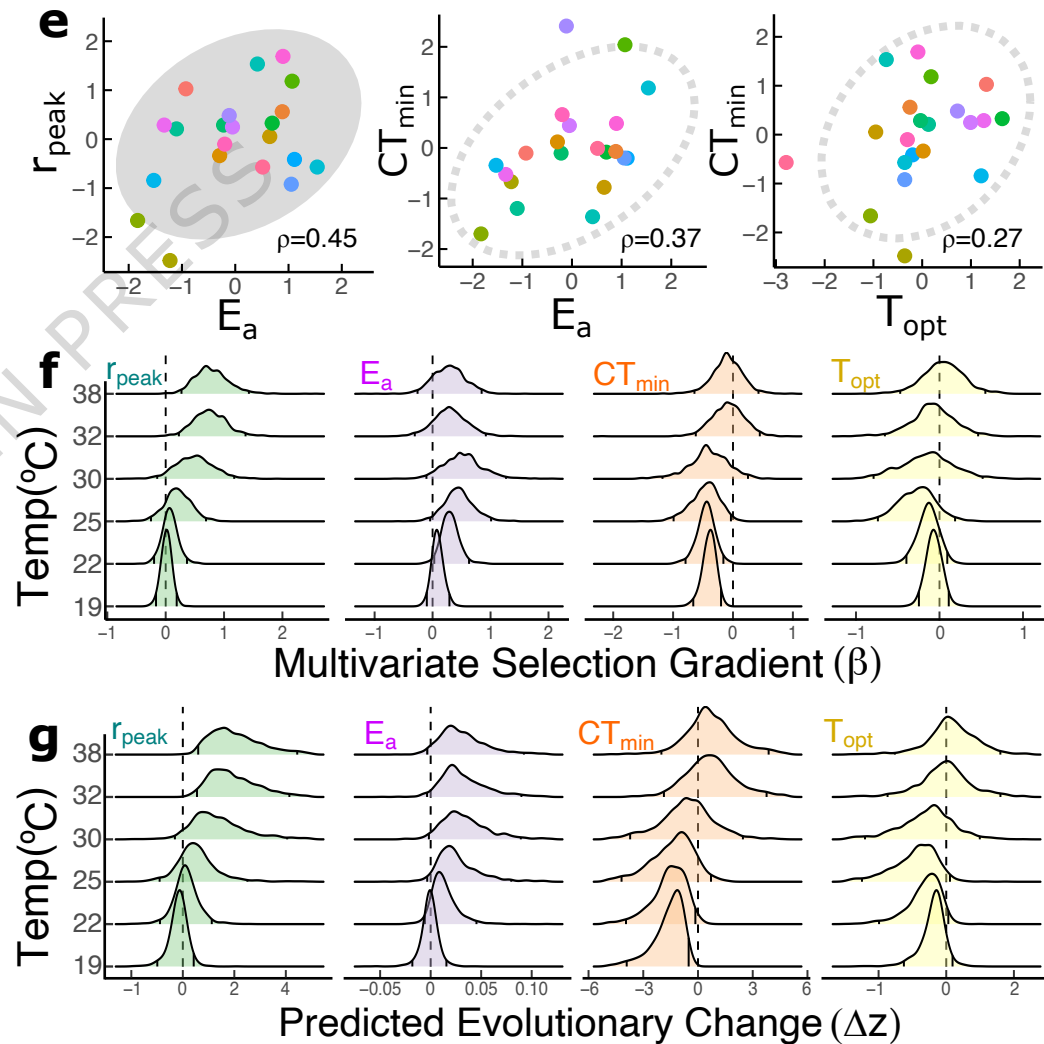
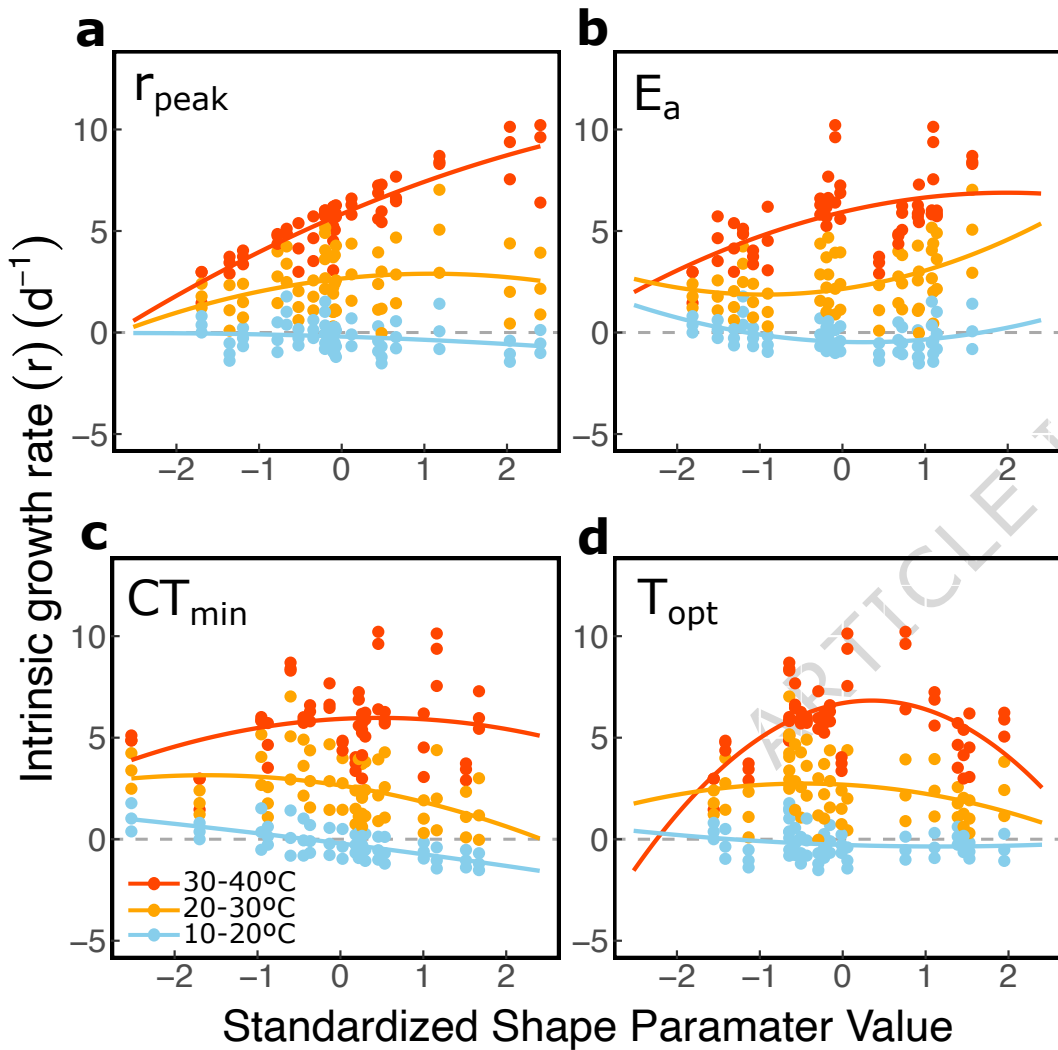
Temperature

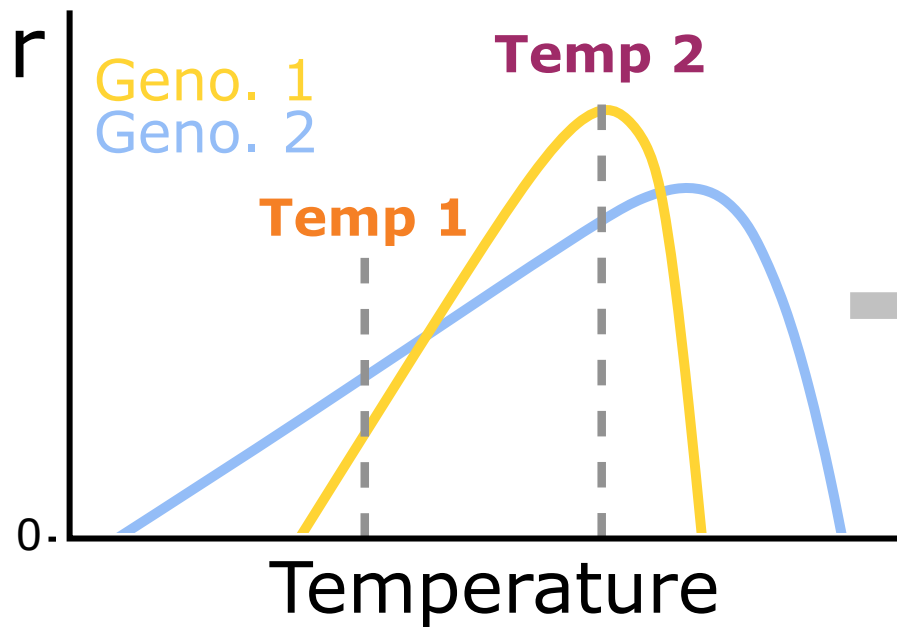
e. Variation in shape parameters

Temperature

aIntrinsic growth rate (r) (d^{-1})**b**





a GxE in r-TPCs**b** Differential selection across temperatures